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(Pages : 2)

Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
MARCH 2025**

Physics/Applied Physics

PHY6B10/APH 6B 10—THERMODYNAMICS

(Admissions Year—2019 Onwards)

Time : Two Hours

Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings.***Section A (Short Answer type)***Answer **all** questions in two **or** three sentences, each correct answer carries a maximum of 2 marks.*

1. Derive the first and second TdS equations.
2. What is meant by a thermodynamic system ?
3. Is it possible to get a Carnot's engine with 100 % efficiency ? Explain.
4. State and explain the zeroth law of thermodynamics.
5. What is a refrigerator and define its co-efficient of performance ?
6. Compare the slopes of adiabatic and isothermals ?
7. What do you mean by an indicator diagram ? Explain it.
8. What is the significance of PV diagram ?
9. State Kelvin-Planck and Clausius statement of Second law of thermodynamics ?
10. What do you mean by quasi static process ?
11. Which are the macroscopic quantities, required to describe the materials in a cylinder of an automobile engine ?
12. Distinguish between reversible and irreversible processes.

(Ceiling = 20 marks)

Turn over

Section B - (Paragraph / Problem Type)

Answer **all** questions in a paragraph of about **half a page to one page**, each correct answer carries a maximum of 5 marks).

13. Deduce the second latent heat equation of Clausius $C_2 - C_1 = \left(\frac{dL}{dT}\right) - \left(\frac{L}{T}\right)$ where C_1 and C_2 represent the specific heat of a liquid and its saturated vapour and L is the latent heat of the vapour.
14. A Carnot's engine whose temperature of the source is 400 K takes 200 calories of heat at this temperature and rejects 150 calories of heat to the sink. What is the temperature of the sink? Also calculate the efficiency of the engine.
15. Show that during a reversible adiabatic process, the entropy of the system remains constant.
16. 1 kg of water at 273 K is brought in contact with a heat reservoir at 373 K. What is the change in entropy of water as it reaches 373 K?
17. Calculate the change in temperature of boiling water when the pressure is increased by 27.12 mm. of Hg. The normal boiling point of water at atmospheric pressure is 100°C. Latent heat of steam = 537 cal/g and specific volume of steam = 1674.
18. Using Maxwell's thermodynamic relations, prove that for any substance, the ratio of the adiabatic and isothermal elasticities is equal to the ratio of the two specific heats.
19. Air at N.T.P is compressed to half of its volume. Calculate the change in its temperature.

(Ceiling = 30 marks)

Section C (Essay Type)

Essays - Answer in about **two pages**, any **one** question.

The question carries 10 marks.

20. Discuss first order phase transition and derive the Clausius-Clapeyron equation.
21. What are isothermal and adiabatic processes? Derive the equation for work done in isothermal and adiabatic processes?

(1 × 10 = 10 marks)